The Bombay Salesian Society's
Don Bosco Institute of Technology
An Autonomous Institute Affiliated to University of Mumbai
Department of Computer Engineering

PROGRESS REVIEW EVALUATION PRESENTATION

Track No. 7 Group No. 5

Centralized Project Proposal & Progress Tracking Portal

SE Mini Project-I (Semester III) | Academic Year 2026–27

Class: SE COMPS C (Batch A)

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Project Overview: Introduction

DBIT ProjectTrack is a centralized web portal that automates topic screening, faculty review and milestone tracking for undergraduate project courses.

ACADEMIC LIFECYCLE

Multi-semester projects (CEP, Mini-Projects, Capstone) demand originality, verifiable stage progression and continuous faculty mentoring.

CURRENT BOTTLENECK

40+ groups per semester submit via Google Sheets, WhatsApp and verbal chats. Faculty verify manually, with no historical archive.

THE PROPOSED INTERVENTION

- **Automated real-time NLP pre-screening:** flags duplicate and paraphrased proposals before submission.
- **Merit-based conflict queue:** side-by-side pitch evaluation instead of first-come allocation.
- **Centralized stage-gate tracker:** transparent progress and permanent written audit trail.

One portal. Three roles.

ProjectTrack

Centralized Proposal & Progress Portal

Student

Faculty Guide

Project Coordinator

SBERT + TF-IDF • Cosine ≥ 0.75 • 3-tier web architecture

Built for SE/TE project courses

Project Overview: Motivation

Informal Channels → **Topic Overlap** → **Review Fatigue** → **Lost Records & Confusion**

WHY PROJECTTRACK?

- First-come allocation lets faster teams claim generic topics and penalises original ones.
- Guides manually cross-check 40+ proposals against past cohorts.
- Review slots go to topic disputes instead of technical mentoring.
- WhatsApp and verbal remarks leave no auditable record.
- Students are unsure of deliverables and next stage gates.

From spreadsheets → structured portal

ProjectTrack combines
DUPLICATE CHECK + MERIT REVIEW + STAGE TRACKING

79.2%	81.8%	44.2%
faced topic overlap	confused on stage gates	waited ≥3 days

Project Overview: Empathy Map and Persona

SAYS

- "I don't have time to cross-check every submission by hand."
- "Tell me which topics actually overlap, not just which sound similar."

THINKS

- Worried about being unfair if two teams get similar marks for duplicate ideas.
- Hopes students won't see evaluation as arbitrary or opaque.

MK

DOES

- Manually skims dozens of titles and abstracts during review rush.
- Keeps disconnected spreadsheets to track approved topics.

FEELS

- Overwhelmed during high-volume submission deadlines.
- Anxious about accidentally approving two near-duplicate projects.

PERSONA

Prof. Meenal Kulkarni

Project Coordinator | 12 yrs teaching

Background: Coordinates 3rd-year CEP projects; reviews 40+ topic submissions per semester.
Goals: Approve fair, non-duplicate topics quickly; keep the process transparent.
Frustrations: No systematic way to spot overlap; disputes eat review time.
Needs: Automated duplicate detection, pitch record per team, simple decision tracker.

Project Overview: Problem Statement

PROBLEM STATEMENT

"To develop a web-based centralized portal for students and project guides that provides automated detection of duplicate or overlapping project topics, a searchable archive of approved topics, and real-time approval, feedback and stage tracking with deadline reminders, thereby reducing overlapping projects and the delays and confusion caused by slow, scattered and untracked feedback."

The project therefore targets a unified, transparent proposal-to-completion workflow.

Scattered process

Sheets, WhatsApp, verbal chats

↓

Current limitations

No overlap check • slow, untracked feedback

↓

Integrated need

One portal from submission to testing

Zero Overlap

NLP similarity filtering

Instant Transparency

Live stage-gate status

Auditable Feedback

Centralized reviewer log

Project Overview: Objectives & Scope

OBJECTIVES

- Automated NLP duplicate detection: flag overlapping titles and abstracts at submission using SBERT and TF-IDF.
- Side-by-side pitch defense: faculty merit review for contested topics instead of first-come allocation.
- Searchable topic archive: pre-submission visibility into past approved projects.
- Stage-gate progress tracking: real-time milestones from approval through testing.

SCOPE

In scope (current implementation)

- Department-level proposal submission
- Semantic duplicate analysis
- Faculty resolution queue
- Milestone tracking and automated reminder alerts for SE/TE courses

Out of scope (future roadmap)

- Full-document thesis plagiarism scanning (Turnitin-style)
- Automated code repository static analysis
- Multi-institute university deployment

Literature Review

Sr. No	Name of paper	Authors	Year of publication	Methodology	Key Findings	Limitations
1.	Web-Based Centralized Student Project Allocation & Management System	S. Sharma, A. Kumar, P. Deshmukh	2021	RBAC, MySQL relational database, PHP web architecture.	Admin follow-ups cut by 40%; eliminated lost physical forms.	Lacks automated duplicate detection.
2.	Detection of Duplicate Project Proposals Using NLP	R. Patel, V. Shah, M. Mehta	2023	Tokenization, TF-IDF vectorization, Cosine similarity.	88% accuracy in identifying exact and near-exact duplicates.	High false-negative rate on rephrased titles.
3.	Semantic Similarity & Paraphrase Detection for Proposals using BERT	J. Dev, S. Nair, L. Verma	2024	Sentence-BERT (SBERT) contextual embeddings, threshold ≥0.75.	94.2% accuracy in detecting paraphrased project titles.	Higher computational overhead.
4.	Expert-in-the-Loop Framework for Academic Conflict Resolution	M. R. Gupta, P. K. Singh	2022	Threshold escalation queue, side-by-side pitch matrix.	Human-in-the-loop review maintained fairness & lowered load.	Dependent on faculty response times.
5.	Digital Archiving & Semantic Search for Student Engineering Projects	A. Joshi, R. Kulkarni	2020	Metadata tag indexing, domain-level relational search.	Archive access reduced redundant topic submissions by 60%.	Static repository without live checks.

Literature Review: Research Gaps

GAP 1: KEYWORD MATCHING FAILS ON PARAPHRASED PROPOSALS

Traditional TF-IDF systems fail when students reword duplicate ideas with synonyms.

PROPOSED SOLUTION

Deploy Sentence-BERT contextual embeddings to detect semantic meaning regardless of phrasing.

GAP 2: ARBITRARY AUTO-REJECTIONS CAUSE STUDENT FRICTION

Binary programmatic rejections ignore nuances and discourage authentic student teams.

PROPOSED SOLUTION

Expert-in-the-Loop Conflict Queue: Flag duplicates for side-by-side pitch evaluations by guides.

GAP 3: ISOLATED STATIC ARCHIVES WITHOUT WORKFLOW INTEGRATION

Historical libraries operate in silos, disconnected from live submission and approval gates.

PROPOSED SOLUTION

Directly link pre-submission repository search into the submission form with live originality scoring.

Website/App Comparison

Sr. No	Name of app/website	Features	Missing Features	References to your work
1.	Google Classroom / MS Teams	File submissions, gradebooks, assignment deadlines.	Automated duplicate alerts, semantic title checks, pitch conflict matrix.	Proves general tools cannot prevent topic hoarding or manage duplicate topics.
2.	Trello / Jira	Visual Kanban boards, stage-gate tracking, activity logs.	Academic proposal forms, NLP similarity checks, guide approval workflows.	Direct inspiration for our visual progress dashboard and stage gates.
3.	Turnitin / iThenticate	Plagiarism detection, similarity scoring, text matching.	Instant title checking, pitch defense judging, lightweight speed.	Demonstrates similarity matching utility, but shows need for real-time pre-checks.
4.	Shared Google Sheets	Low-cost collaborative spreadsheet logging.	Automated duplicate alerts, overwrite protection, structured feedback logs.	Represents the flawed baseline causing coordinator burnout and lost records.
5.	GitHub / GitLab Projects	Repository archiving, milestone tracking, PR reviews.	Early proposal screening, semantic topic checks, non-code pitch evaluation.	Validates multi-semester repository model while highlighting early-stage gaps.

Final Feature List

Name of app/website: Centralized Project Proposal & Progress Tracking Portal (DBIT ProjectTrack)		
Sr. No	Features	Justification
1.	Real-Time NLP Duplicate & Similarity Detection	Scans proposed titles and abstracts during submission via SBERT and TF-IDF to flag matching or reworded topics instantly, preventing topic hoarding and manual cross-checking.
2.	Faculty Pitch Defense & Conflict Resolution Dashboard	Provides a side-by-side review interface where faculty evaluate brief pitch defenses from teams with overlapping topics, ensuring allocation is based on merit rather than speed.
3.	Searchable Historical Project Repository	Grants students pre-submission access to a domain-filtered archive of previously approved projects, enabling them to verify topic originality during brainstorming.
4.	Visual Stage-Gate Progress Tracker	Tracks milestones across structured phases (Topic Approval → Design → Dev → Testing) on a shared dashboard, resolving 81.8% of student confusion regarding next deliverables.
5.	Centralized Written Feedback Log & Automated Reminders	Permanently stores faculty review comments and sign-off stamps while sending automated reminder alerts for pending reviews, resolving feedback stagnation.

Survey & Proposed Solution

Survey Data & Interpretation (n = 79)

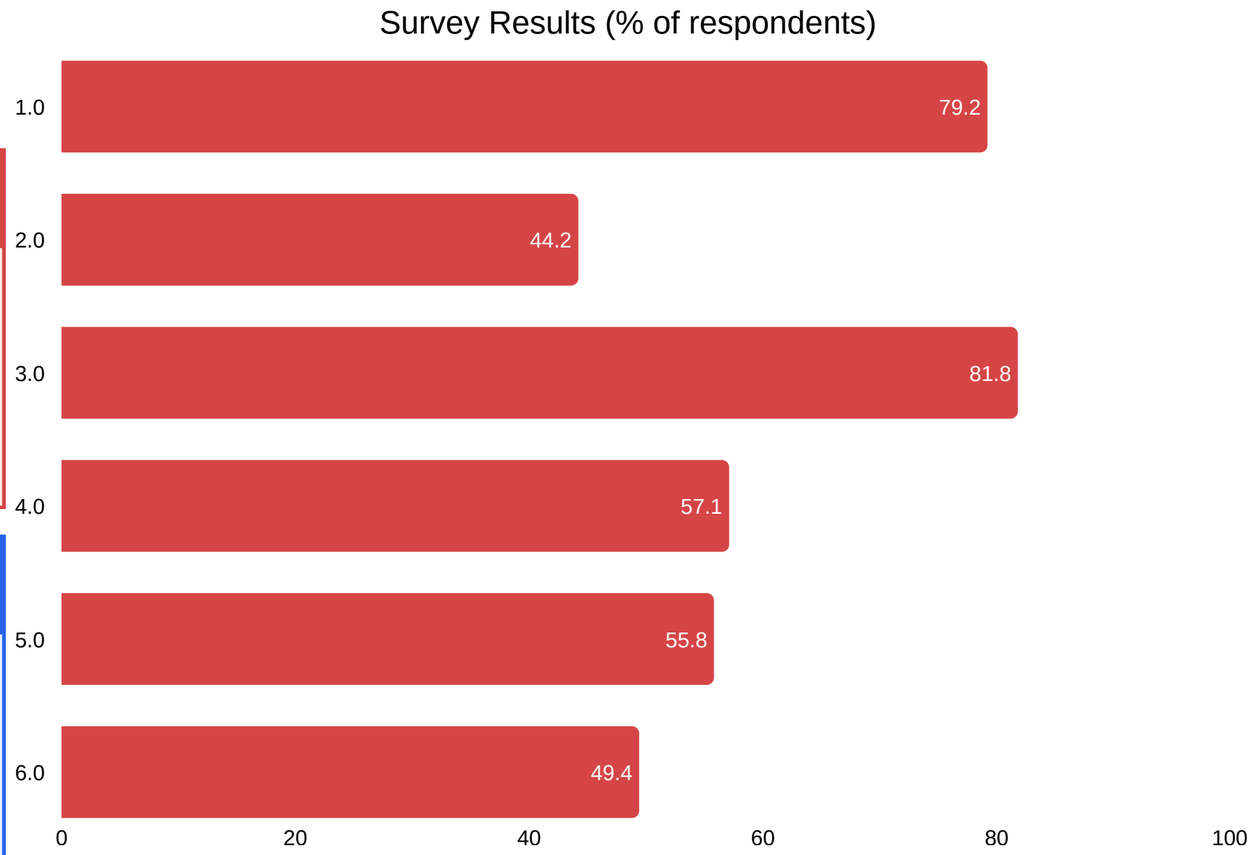
Google Forms survey of 77 SE/TE students and 2 project guides across DBIT Computer Engineering cohorts.

Key Findings

- **Topic Overlap Encountered:** 79.2% (61/77)
- **Wait Time ≥3 Days for Feedback:** 44.2% (34/77)
- **Confusion on Next Stage Gates:** 81.8% (63/77)

Top Student Feature Demands

- Duplicate Check (57.1%)
- Topic Search (55.8%)
- Alerts & Reminders (49.4%)



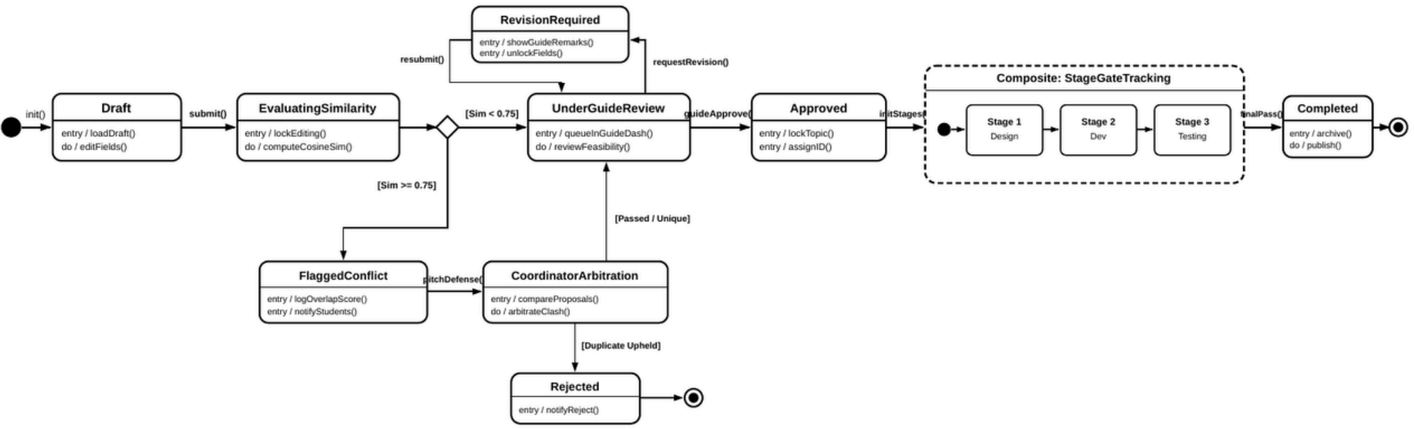
Survey & Proposed Solution

Proposed System Architecture

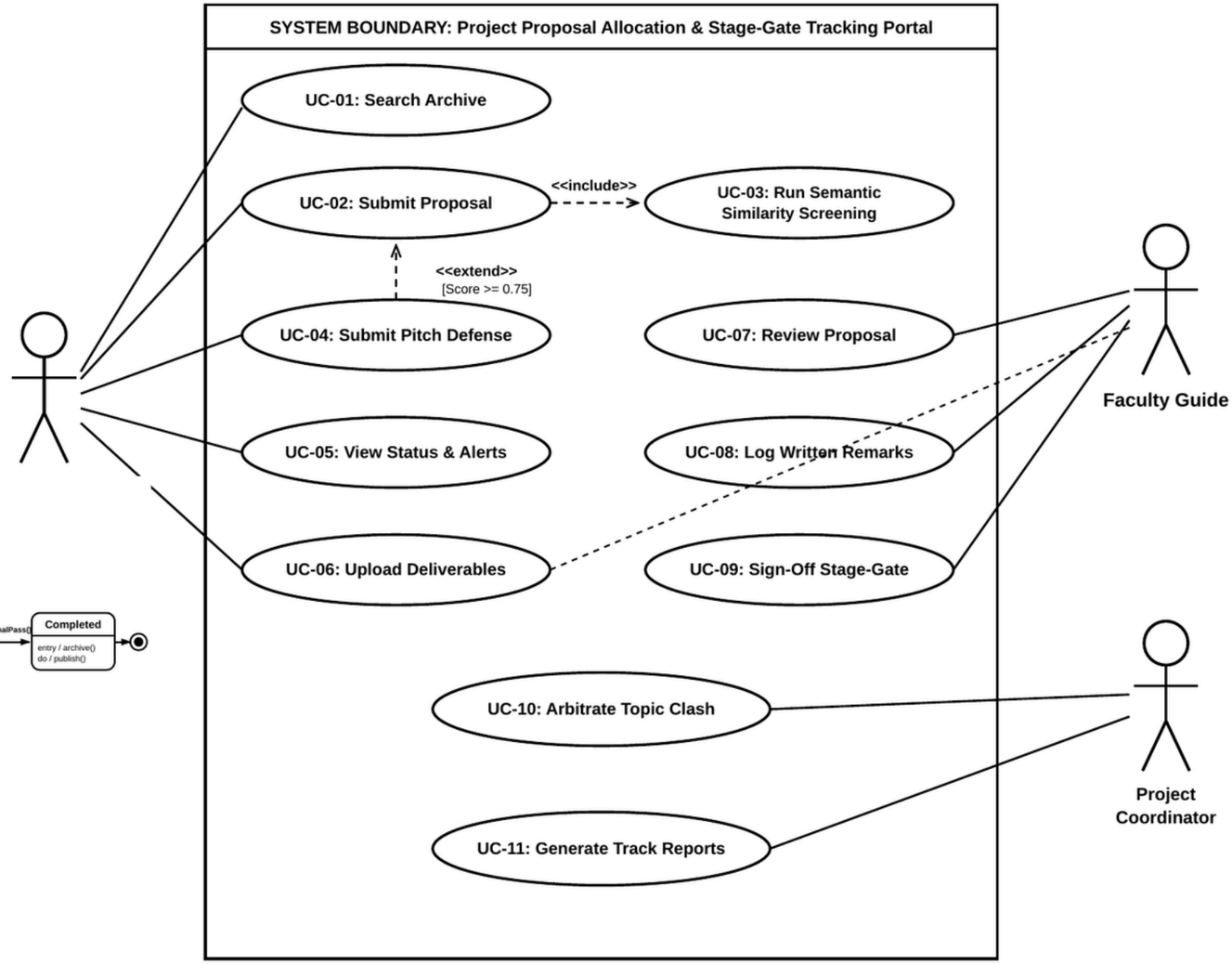
3-Tier Pipeline: Student/Faculty Web Portal ↔ NLP Similarity Engine ↔ Historical Relational DB.

UML Use Case & Activity Workflow & UML State Machine Lifecycle

UML 2.5 State Machine Diagram: Project Proposal & Stage-Gate Lifecycle



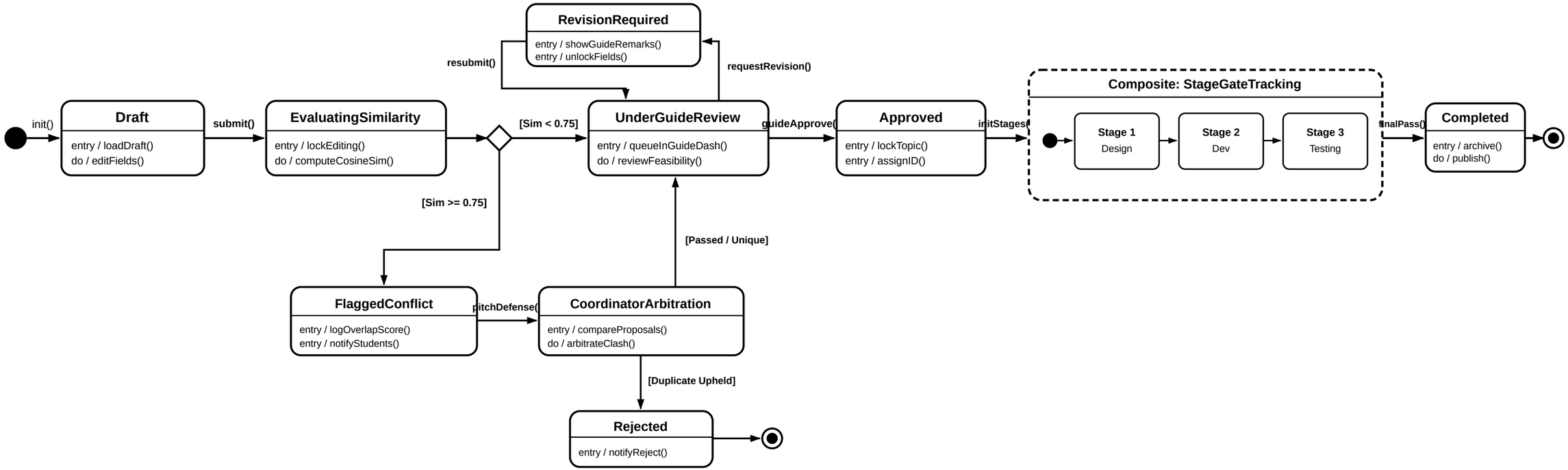
UML Use Case Diagram: Project Proposal Allocation & Tracking Portal



Survey & Proposed Solution

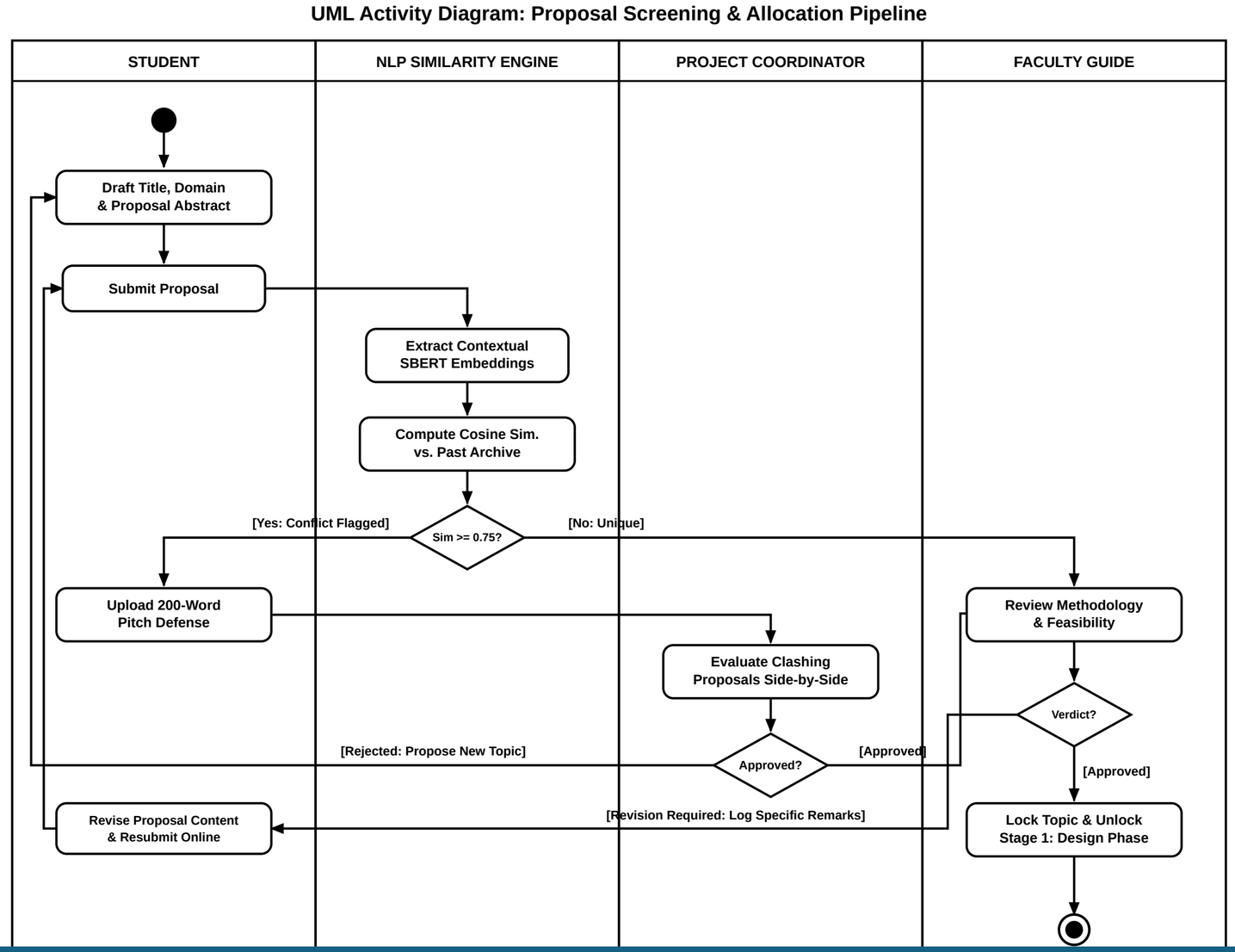
UML STATE MACHINE LIFECYCLE

UML 2.5 State Machine Diagram: Project Proposal & Stage-Gate Lifecycle



Survey & Proposed Solution

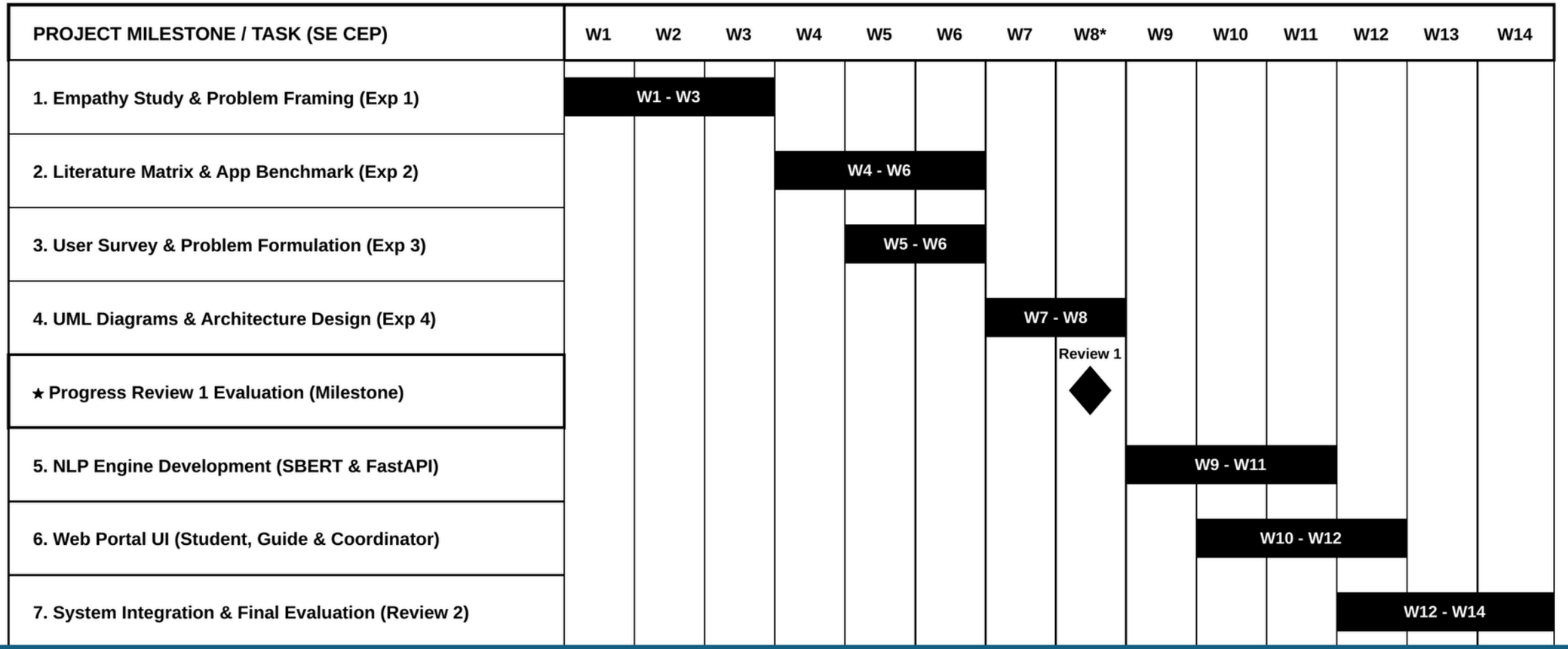
UML ACTIVITY DIAGRAM



Survey & Proposed Solution

Project Timeline & Gantt Chart

Project Implementation Schedule: 14-Week Milestone Gantt Chart



Proposed Solution

1 Submit

Team enters title and abstract in the web portal

2 Semantic Check

SBERT + TF-IDF score similarity against past and live proposals

3 Faculty Review

Flagged duplicates go to a side-by-side pitch review

4 Track & Remind

Stage-gate milestones with automated alerts

Semantic Duplicate Detection

- Sentence-BERT embeddings catch reworded ideas
- TF-IDF cosine similarity as a baseline
- Live originality score on the submission form

Expert-in-the-Loop Queue

- No arbitrary auto-rejection
- Guides compare pitches side by side
- Decision and feedback logged for audit

Integrated Workflow

- Searchable archive of approved projects
- Stage-gate tracker for SE/TE courses
- Role-based access: student, guide, coordinator

References

Sr. No	References
1.	S. Sharma, A. Kumar, P. Deshmukh (2021). "Web-Based Centralized Student Project Allocation and Management System." International Journal of Computer Applications.
2.	R. Patel, V. Shah, M. Mehta (2023). "Detection of Duplicate Project Proposals Using Natural Language Processing." IEEE Access, vol. 11, pp. 45120–45129.
3.	J. Dev, S. Nair, L. Verma (2024). "Semantic Similarity and Paraphrase Detection for Academic Proposals using SBERT." ACM Transactions on Computing Education.
4.	M. R. Gupta, P. K. Singh (2022). "Expert-in-the-Loop Framework for Academic Conflict Resolution and Allocation." Journal of Educational Technology Systems, vol. 51.
5.	A. Joshi, R. Kulkarni (2020). "Digital Archiving and Semantic Search for Historical Student Engineering Projects." Springer CCIS, vol. 1324.
6.	Visual Paradigm & OMG UML 2.5 Specification (2024). UML Modeling Reference Guides (Use Case, Activity, State Machine). https://www.visual-paradigm.com/guide/uml-unified-modeling-language/

THANK YOU

QnA

We invite questions, suggestions, and feedback from the Progress Review Committee.

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